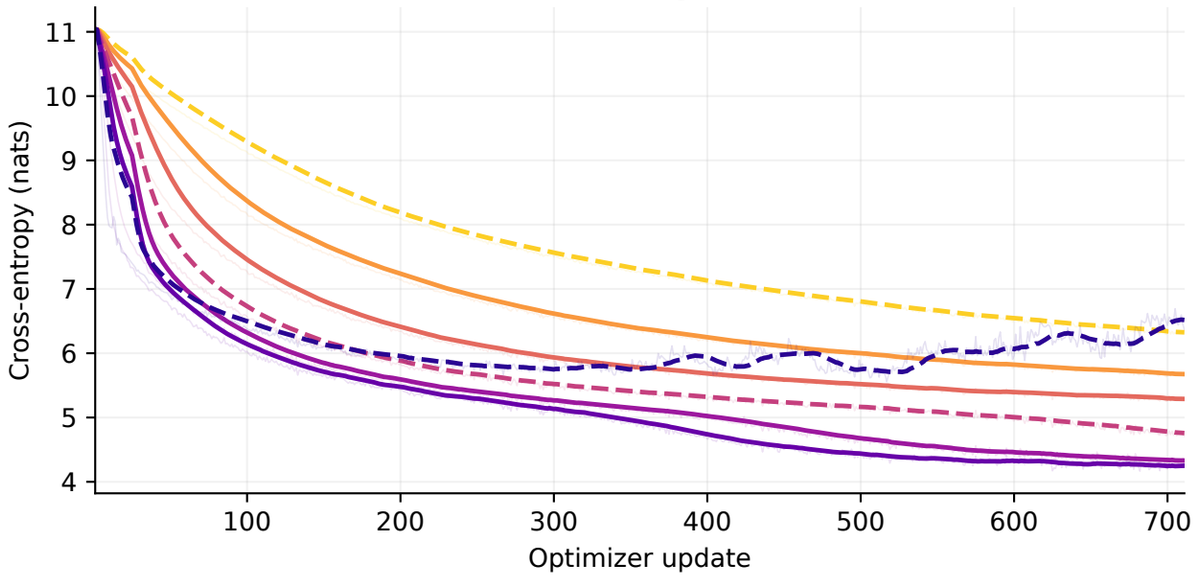
Pythia-14M: seven warmup-only learning rates

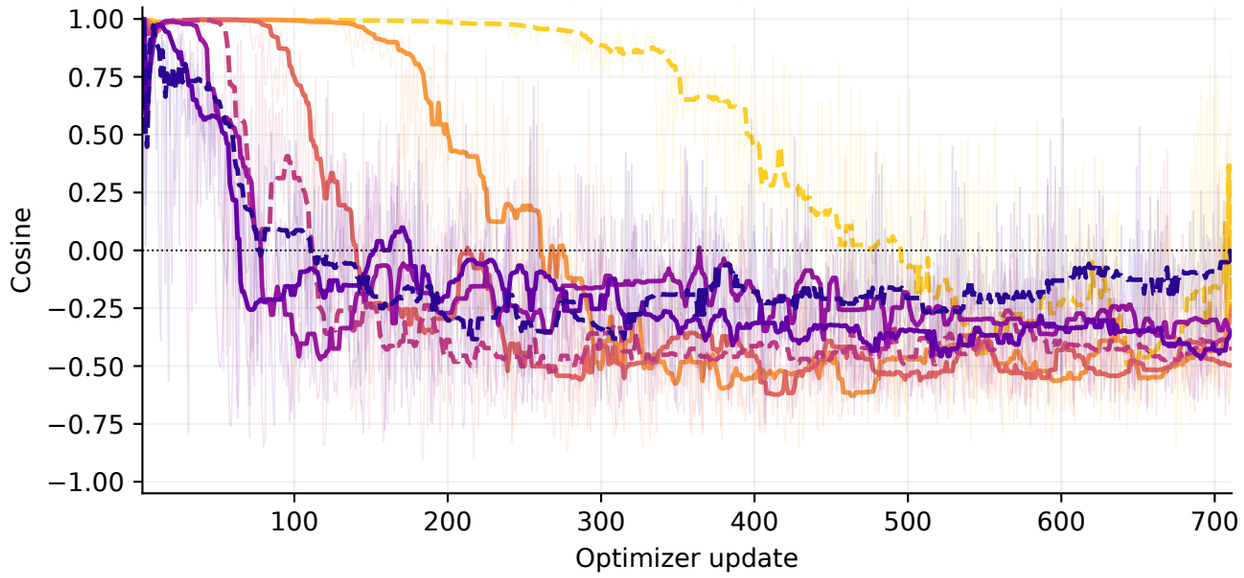
Peak LR: light yellow to dark purple as LR increases; S = Spark, R = RTX4090
Dashed = Spark; solid = RTX4090

0.0001 (S) · 0.000215443 (R) · 0.000464159 (R) · 0.001 (S) · 0.00215443 (R) · 0.00464159 (R) · 0.01 (S)

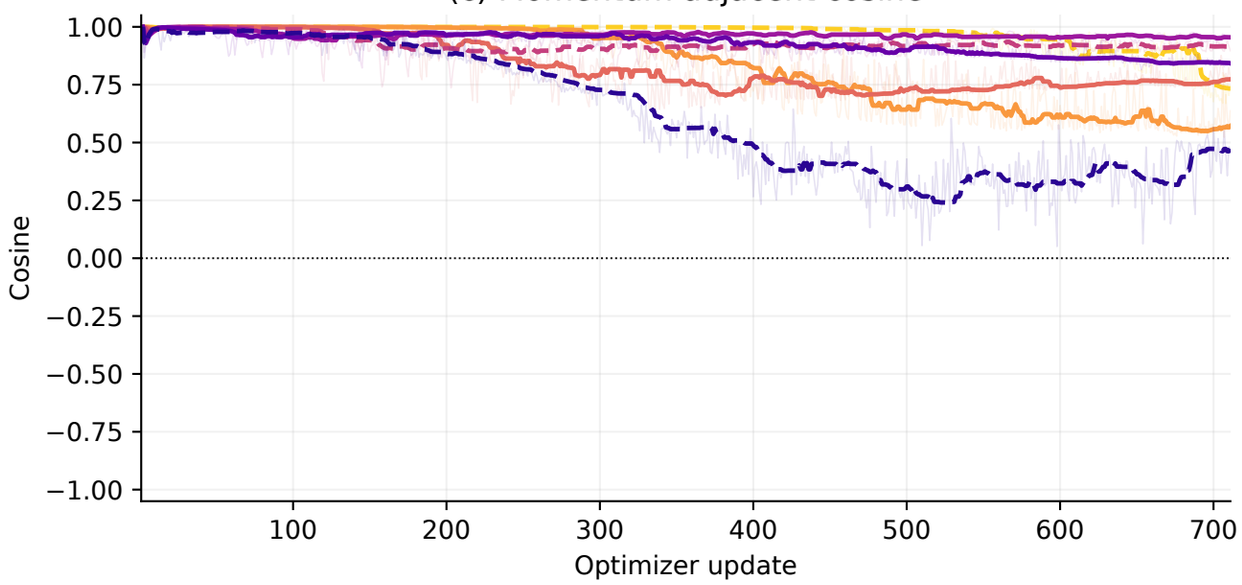
(a) Training loss



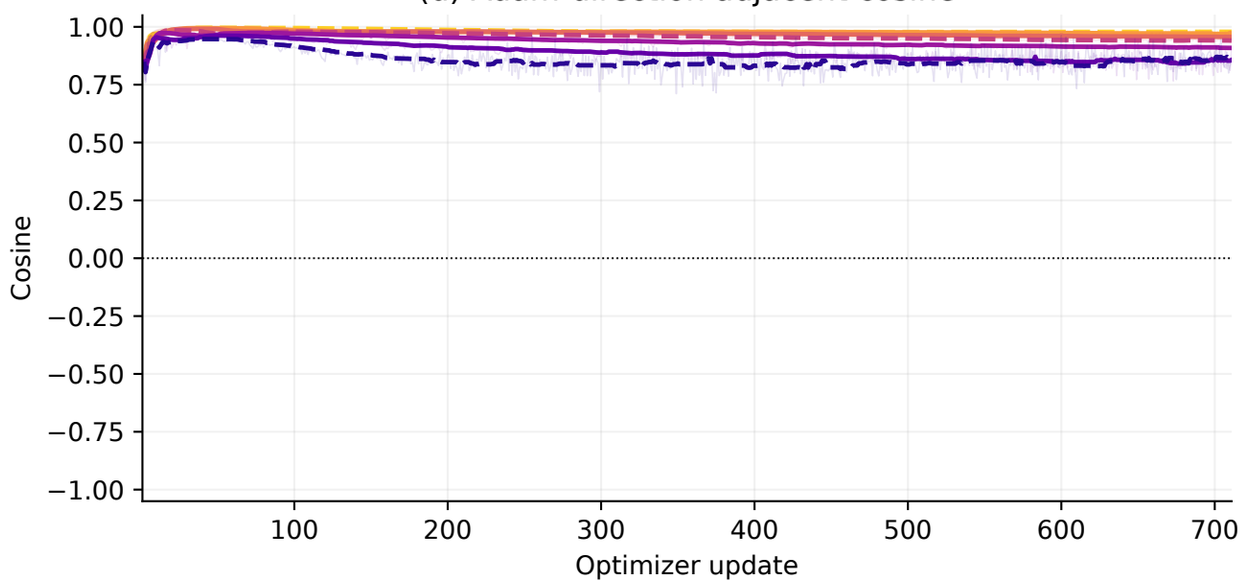
(b) Raw-gradient adjacent cosine



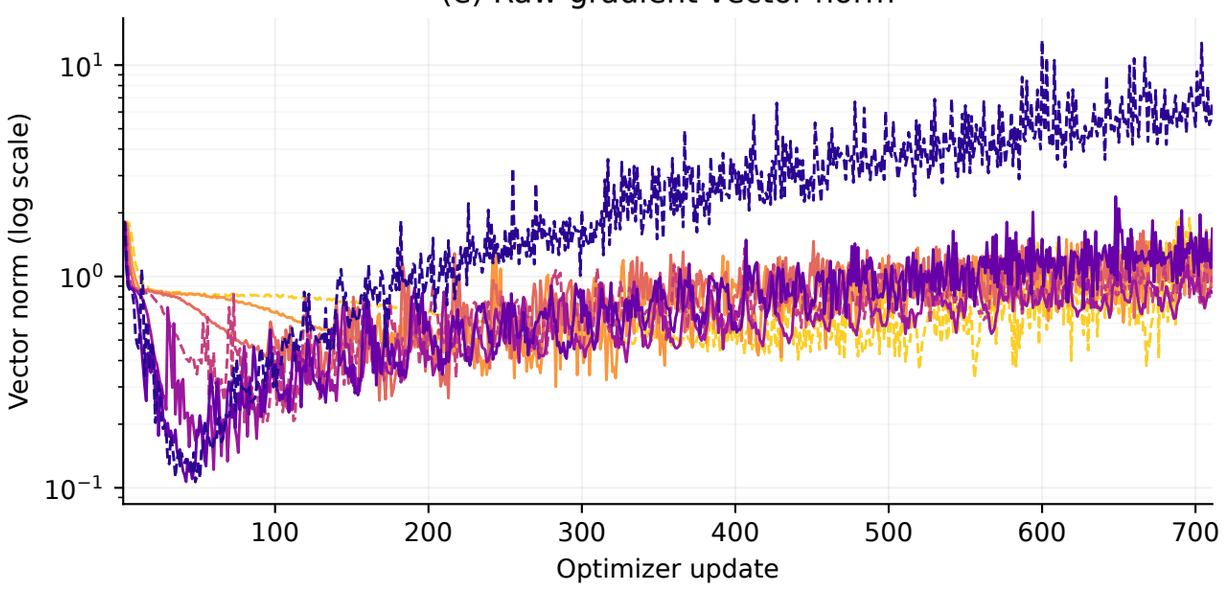
(c) Momentum adjacent cosine



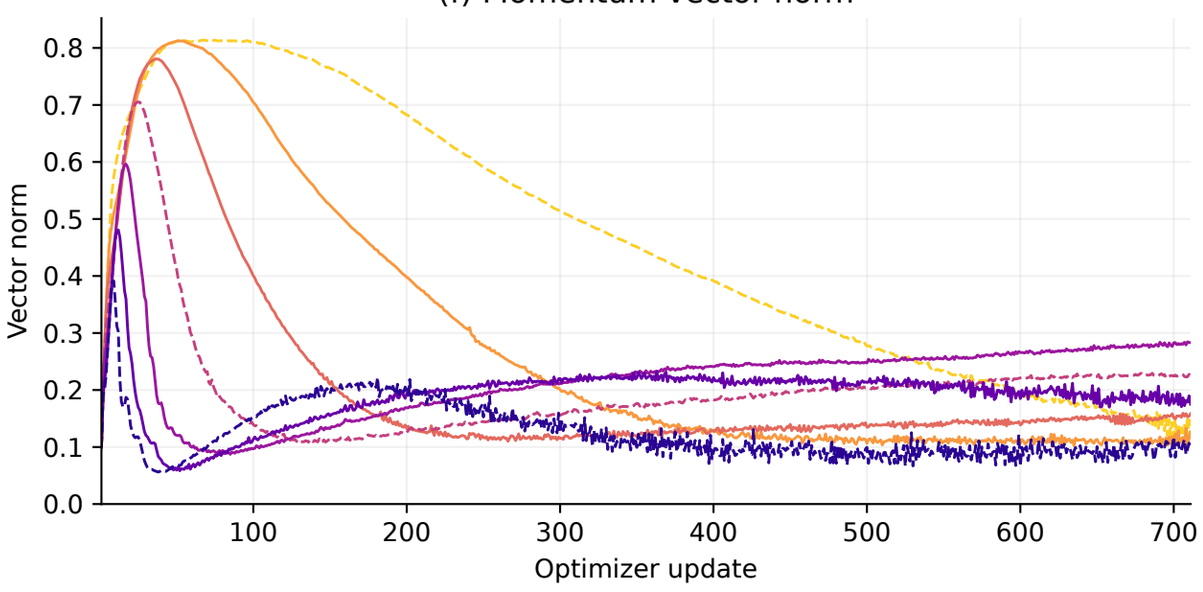
(d) Adam-direction adjacent cosine



(e) Raw-gradient vector norm



(f) Momentum vector norm



Light traces: updates; bold loss: trailing 25-update token-weighted mean; bold cosine: trailing 25-update median.
Norms are unsmoothed: panel (e) logarithmic Y, panel (f) linear Y. Display 1-711; partial update 712 retained in results.
Same frozen initialization/order, seed 1234. Different hardware across S/R; one seed; LR also scales AdamW decay.